THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 13: **Our Environment** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An ecosystem consists of 1

(a) only plants **(b)** only animals

(c) only non-living things **(d)** biotic and abiotic components

2. A food chain always begins with 1

(a) herbivores **(b)** decomposers **(c)** producers **(d)** carnivores

3. The ozone layer is present in the 1

(a) stratosphere **(b)** troposphere **(c)** hydrosphere **(d)** lithosphere

4. Which of the following is an abiotic component of an ecosystem? 1

(a) Temperature **(b)** Grass **(c)** Bacteria **(d)** Fungi

5. Which of the following is a correct food chain? 1

(a) Grass, grasshopper, frog, snake **(b)** Frog, grass, snake, grasshopper

(c) Snake, frog, grasshopper, grass **(d)** Grass, snake, frog, grasshopper

6. Ozone is a molecule made of 1

(a) three oxygen atoms **(b)** two oxygen atoms

(c) oxygen and nitrogen **(d)** one oxygen atom

7. Which of the following are producers? 1

(a) Deer **(b)** Snakes **(c)** Green plants **(d)** Fungi

8. The various steps in a food chain are called 1

(a) niches **(b)** habitats **(c)** trophic levels **(d)** ecosystems

9. The ozone layer protects the earth from 1

(a) visible light **(b)** sound waves

(c) harmful ultraviolet radiation **(d)** infrared rays

10. Organisms that depend on others for food are called 1

(a) decomposers only **(b)** autotrophs **(c)** consumers **(d)** producers

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Differentiate between biotic and abiotic components of an ecosystem. 2

12. What is a trophic level? 2

13. What is the role of the ozone layer? 2

14. What are producers? Give two examples. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the difference between a food chain and a food web. 3

16. State three reasons for the role of decomposers in the ecosystem. 3

17. Differentiate between a natural and an artificial ecosystem with examples. 3

OR

Describe the flow of energy in an ecosystem. 3

18. What would happen if all the decomposers were removed from the ecosystem? 3

OR

Explain how the biomagnification of pesticides occurs. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. What is an ecosystem? Describe its biotic and abiotic components with examples. 5

OR

What is biomagnification? Explain with an example of a food chain how a non-degradable pesticide gets concentrated. 5

20. Explain the different types of organisms in an ecosystem (producers, consumers, decomposers) and their roles with examples. 5

OR

Distinguish between biodegradable and non-biodegradable substances with examples. Explain the problem each poses when thrown away. 5

END OF QUESTION PAPER — Total: 40 Marks

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